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[Diagnostic Test Accuracy Protocol]**Artificial intelligence for glaucoma**

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ABSTRACT**Objectives**

This is a protocol for a Cochrane Review (diagnostic). The objectives are as follows:

To determine the accuracy of artificial intelligence (AI) algorithms as a diagnostic tool for glaucoma compared with human graders in a community or secondary care setting.

Secondary objectives

1. To compare the performance of different AI algorithms in the diagnosis of glaucoma
2. To explore other potential causes of heterogeneity in the diagnostic performance compared with human graders, using subgroup analysis of the following characteristics:
 - a. Clinical setting in which the test is used (the general population in the community versus people referred to secondary care);
 - b. Study design (studies enrolling consecutive participants in the same setting versus multicentre registries and public databases);
 - c. Characteristics of the population (age according to quartiles, sex, symptomatic versus asymptomatic), when sufficient data are available;
 - d. Prevalence of glaucoma in the training and test sets (< 5% versus ≥ 5%), since sensitivity and specificity may depend on disease prevalence [17];
 - e. Severity of glaucoma;
 - f. Core AI method used (neural networks, random forests, support vector machines, or others);
 - g. Size of the dataset from which performance data were collected (< 1000 versus ≥ 1000 total unique participants);
 - h. Modalities of input data for the AI algorithms (imaging data, visual fields, clinical parameters, or any combination), particularly as their accessibility and affordability may vary across different settings.

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BACKGROUND

Diagnosis of glaucoma traditionally relies on a comprehensive evaluation of the optic nerve head (ONH) and retinal nerve fibre layer (RNFL) through retinal examinations and imaging tools, visual field (VF) assessments, and intraocular pressure (IOP) measurements [1]. However, with the traditional case-finding pathway, difficulties in detecting glaucoma early coexist with high false positive rates, which increase the burden on healthcare systems. In secondary care, where the prevalence of glaucoma is higher, avoiding false negatives is important. These challenges not only contribute to potential vision loss if disease is not addressed promptly, but also highlight the inefficiencies within the conventional diagnostic framework. Specifically, the current approach to glaucoma detection is notably time-consuming and heavily dependent on clinicians' expertise and experience, which can vary widely [2]. Implementing artificial intelligence (AI) in clinical settings may revolutionise this process, by enhancing diagnostic accuracy and reducing the time people spend in clinics, ultimately leading to more efficient and reliable glaucoma diagnosis and management.

Increasingly, AI is being developed for diagnostic tools and treatment algorithms in clinical settings. For example, AI has been used in the diagnosis of cancer, using medical imaging and classification algorithms [3, 4]. In ophthalmology, significant developments within the field of retinal disease have placed ophthalmology at the forefront of AI [5, 6]. In glaucoma, much work is being done to investigate how AI can facilitate the disease detection and monitor its progression [7, 8].

The purpose of this systematic review is to determine whether AI is currently a valid tool to diagnose glaucoma in a community or secondary care setting.

Target condition being diagnosed

Glaucoma is a group of optic neuropathies, characterised by progressive degeneration of the retinal ganglion cell, resulting

in cupping – a characteristic appearance of the optic disc – and visual loss. Glaucoma is often called the silent thief of sight. Approximately 10% of the 70 million people with glaucoma are bilaterally blind, making it the leading cause of irreversible blindness in the world [9, 10]. Glaucoma is classified into two broad categories: open-angle and closed-angle. Both open-angle and closed-angle glaucoma can occur without a known underlying cause, as primary glaucoma. Secondary glaucoma can result from trauma, medication, inflammation, tumour, and other conditions (e.g. pigment dispersion or pseudoexfoliation).

Index test(s)

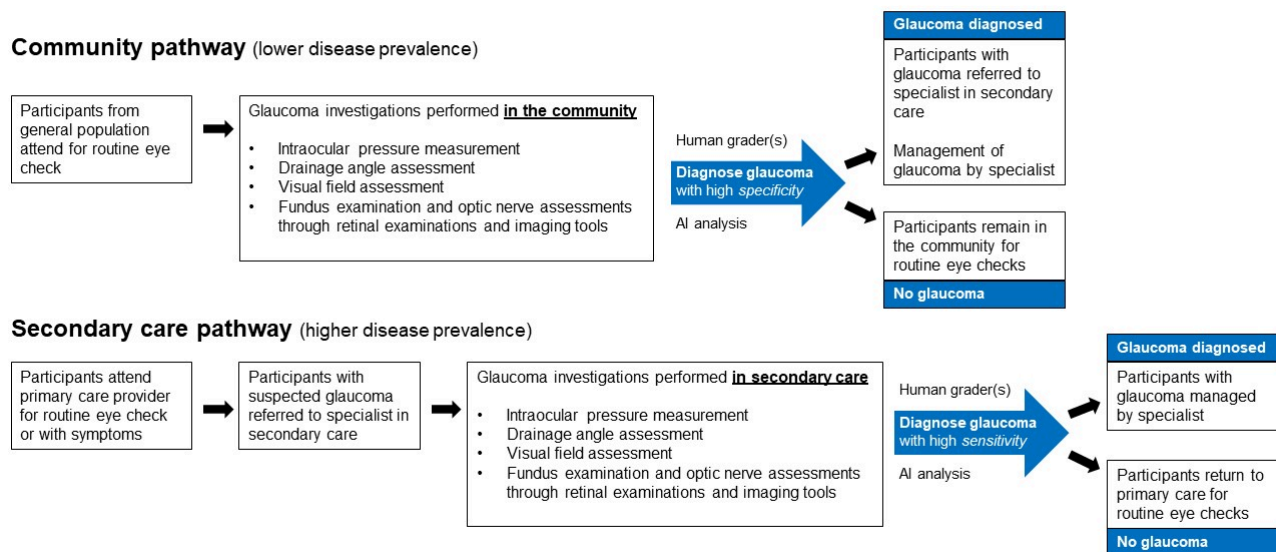
Artificial intelligence (AI) is a field within computer science that focusses on performing tasks that typically require human intelligence. Machine learning (ML) is a subset of AI that involves training a machine to learn from data to make decisions. Deep learning, a further subset of ML, uses complex algorithms, called neural networks, to process data in layers, enabling the machine to improve its tasks through experience and large amounts of data.

AI algorithms may be applied to a diverse array of input data modalities, including ocular examination parameters, imaging tests, such as colour fundus photographs and optical coherence tomograms, and the results of psychophysical tests, such as automated perimetry.

For this review, we will define the index test as an AI algorithm that processes any form of input data for diagnosing glaucoma. The scope of this definition is intentionally broad, because there is no existing consensus about which AI algorithms and input data demonstrate optimal performance in diagnosing glaucoma.

Clinical pathway

There are two anticipated clinical pathways, outlined in [Figure 1](#), because of the differences in how glaucoma pathways could be set up in different healthcare systems.

Figure 1. Clinical pathways for participants seeking care for glaucoma

The first pathway, the 'community' pathway, involves participants from the general population who do not have visual symptoms, and who are not necessarily suspected to have glaucoma. Participants have glaucoma investigations carried out in the community, for example by an optometrist. An AI algorithm analyses the data to diagnose or exclude glaucoma. Participants who are found to have glaucoma are referred to a glaucoma specialist in secondary care for management. Those in whom glaucoma is excluded remain in the community for routine eye checks.

The second clinical pathway, the 'secondary care' pathway, starts when a participant presents to their primary care provider (e.g. primary care doctor, community optician, or general ophthalmology clinic) with either new visual symptoms or for a routine eye check. If there is a suspicion of glaucoma, they are referred to a glaucoma specialist in secondary care for glaucoma investigations. An AI algorithm analyses the data to diagnose or exclude glaucoma. Those who are diagnosed with glaucoma remain in secondary care for management by a glaucoma specialist; those in whom glaucoma is excluded return to their primary care provider for routine eye checks.

The expected disease prevalence at the point of glaucoma investigations should be higher in the 'secondary care' pathway than the 'community' pathway. Specificity would need to be high in the 'community' pathway, because false positive diagnoses would result in unnecessary referrals to secondary care, creating an unwanted burden on this service. By contrast, sensitivity would need to be high in the 'secondary care' pathway, because the expected disease prevalence is relatively high, and false negative diagnoses would likely cause loss of vision.

Rationale

Comprehensively assessing the precision of AI in diagnosing glaucoma bears immense significance, potentially influencing both clinical protocols, and broader public health strategies. The advent of AI algorithms to accurately detect glaucoma in low prevalence settings might pave the way for community-based glaucoma screening programmes — an area currently lacking due to the absence of tests that combine both high specificity and cost-effectiveness. In addition, and more importantly, should AI algorithms demonstrate robust diagnostic accuracy, they could revolutionise secondary care by expediting the triage process, thereby, liberating specialists to focus on managing people with glaucoma. However, previous studies evaluating AI in glaucoma diagnostic pathways have been limited by their focus on specific input modalities, or the lack of meta-analysis of the results [11, 12, 13, 14].

This systematic review will also complement other Cochrane reviews exploring the accuracy of AI in detecting ocular diseases [15, 16].

OBJECTIVES

To determine the accuracy of artificial intelligence (AI) algorithms as a diagnostic tool for glaucoma compared with human graders in a community or secondary care setting.

Secondary objectives

1. To compare the performance of different AI algorithms in the diagnosis of glaucoma
2. To explore other potential causes of heterogeneity in the diagnostic performance compared with human graders, using subgroup analysis of the following characteristics:

- a. Clinical setting in which the test is used (the general population in the community versus people referred to secondary care);
- b. Study design (studies enrolling consecutive participants in the same setting versus multicentre registries and public databases);
- c. Characteristics of the population (age according to quartiles, sex, symptomatic versus asymptomatic), when sufficient data are available;
- d. Prevalence of glaucoma in the training and test sets ($< 5\%$ versus $\geq 5\%$), since sensitivity and specificity may depend on disease prevalence [17];
- e. Severity of glaucoma;
- f. Core AI method used (neural networks, random forests, support vector machines, or others);
- g. Size of the dataset from which performance data were collected (< 1000 versus ≥ 1000 total unique participants);
- h. Modalities of input data for the AI algorithms (imaging data, visual fields, clinical parameters, or any combination), particularly as their accessibility and affordability may vary across different settings.

METHODS

Criteria for considering studies for this review

Types of studies

We will include studies evaluating the diagnostic performance of one or more artificial intelligence (AI)-based index tests against the reference standard. Studies must report single test or comparative diagnostic test accuracies based upon enrolment of participants in the same setting, or studies based upon large-scale multicentre registries and public databases. All participants must have received the index test(s) followed by the reference standard, except in randomised comparative accuracy studies, in which study participants are randomly allocated to receive one of two or more index tests followed by the reference standard.

We will exclude diagnostic case-control studies (two-gate design) because they may overestimate sensitivity and specificity [18].

For comparative test accuracy studies reporting the accuracy of two or more index tests against the reference standard, we will include paired accuracy studies (in which all participants receive all index tests) and randomised comparative accuracy studies (in which participants are randomly allocated to receive one index test). We will exclude non-randomised comparative accuracy studies, in which participants are allocated to receive one of the index tests only, without randomisation, because such studies are at risk of selection bias and may suffer from lack of representativeness.

Participants

We will include studies investigating participants of all ages, both symptomatic and non-symptomatic, regardless of whether they actively sought eye care.

To focus on studies that evaluated the performance of AI algorithms as a diagnostic test, we will exclude studies in which participants were previously diagnosed with glaucoma, or have started treatment for glaucoma.

Index tests

We will only consider autonomous diagnostic AI algorithms that were either internally or externally validated. For internal validation, we will include AI algorithms that went beyond the initial training stage, and underwent an internal validation or testing phase (e.g. cross-validation or hold-out). Because some publications use 'validation' and 'testing' interchangeably, we will define a 'test set' (or 'validation set' or 'hold-out set') as one that is reserved, or held back, for the final evaluation of the AI algorithm, whether it is a portion of the original participant sample ('internal') or a distinct sample of participants ('external'). We will also accept studies that used 'cross-validation' techniques, in which several splits, or 'folds', of the test dataset were generated, and performance results were averaged across the splits.

Examples of relevant AI methods include neural networks, random forests, and support vector machines [19, 20]. We will include algorithms that accept any modality or combination of input material, including, but not limited to, optical coherence tomography (OCT) and colour fundus photographs, visual field test results, and clinical parameters. For OCT, we will include algorithms that utilise both images and numeric variables derived from OCT raw data. OCT images may be obtained from different areas, such as from the peripapillary nerve fibre layer, optic nerve head, or macular region, and may be segmented or unsegmented, or extracted as geometric models. For visual field tests, we will include algorithms that utilise both images obtained from a visual field (VF) test printout and raw data. We will classify the index tests based upon the input data they utilise: (i) OCT alone or colour fundus photography alone, or both together, (ii) visual field tests alone, (iii) clinical variables alone, or (iv) any combination.

Target conditions

The target condition is glaucoma. We will include both primary and secondary glaucoma, and open- and closed-angle forms of the disease. When studies report accuracy for differentiating glaucoma of multiple severity levels, we will only extract test performance data for the diagnosis of glaucoma. We will include studies that report the detection of suspected glaucoma ('glaucoma suspect'). If glaucoma and glaucoma suspects are classified separately, we will combine these categories, because both require further specialist evaluation or monitoring.

Reference standards

Typically, a clinical diagnosis of glaucoma is based upon the following criteria [21].

1. Visual field assessment using standard automated perimetry
2. Fundus examination and assessment of the optic nerve, using stereoscopic slit lamp biomicroscopy, with pupil dilatation
3. Measurement of intraocular pressure using tonometry
4. Gonioscopy to assess peripheral anterior chamber configuration and depth

We will include studies in which the reference standard is established by one or more glaucoma specialists, ophthalmologists, or optometrists with experience and training in managing people with glaucoma, as reported by the study authors.

Search methods for identification of studies

Electronic searches

We will search:

1. the Cochrane Central Register of Controlled Trials (CENTRAL; which contains the Cochrane Eyes and Vision Trials Register; latest issue), in the Cochrane Library;
2. Ovid MEDLINE, Ovid MEDLINE E-pub Ahead of Print, Ovid MEDLINE In-Process and Other Non-Indexed Citations, Ovid MEDLINE Daily (January 1946 to present);
3. Embase (January 1947 to present);
4. PubMed (1946 to present);
5. Latin American and Caribbean Health Sciences Literature Database (LILACS; 1982 to present);
6. Compendex (1884 to present);
7. US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (www.clinicaltrials.gov; current search date);
8. World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP; www.who.int/ictip/search/en; current search date).

We will not use any date or language restrictions in the electronic search for trials. We will exclude unpublished work, such as preprints, thesis work, conference abstracts, or opinion pieces, such as editorials or letters to the editor.

For details of search strategies, see [Supplementary material 1](#).

Searching other resources

We will search the reference lists of primary included studies, and reviews and meta-analyses identified in the electronic searches, as outlined in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy* [22]. We will search MEDLINE and Retraction Watch [23], to ensure that none of the included studies were withdrawn owing to errors or fraud. If we encounter a retraction notice for an included study, before or at the data extraction stage, we will exclude that study and document the reasons for exclusion.

Data collection and analysis

Selection of studies

We will import records from the database searches into Covidence [24]. Two review authors will independently screen the titles and abstracts. We will classify each study as 'yes' (relevant), 'maybe' (possibly relevant), or 'no' (not relevant). After retrieving full-text copies of all relevant and possibly relevant records, two review authors will independently assess the full-text records to determine their eligibility. If there is a disagreement regarding eligibility, a third review author will arbitrate at each stage. We will contact the study authors for clarification, if necessary.

Data extraction and management

Two review authors will independently extract data from included studies into a standardised form in Covidence [24], developed by Cochrane Eyes and Vision US Project. We will extract the following data from the included studies:

1. Study methods (population sampling): study design, single or multicentre study, enrolment period;

2. Participant characteristics and setting: country and setting of recruitment, inclusion and exclusion criteria, total sample size (participants, eyes, or images), number of participants, eyes, or images with the target condition, characteristics of participants (e.g. age, sex, race, ethnicity, ocular comorbidity), sources and size of the datasets (numbers of participants, eyes, and images);
3. Index test(s): technical aspects (e.g. full name, manufacturer, and version number), threshold(s) for test positivity, details of algorithm development (training, internal validation (testing), and external validation, if performed), procedures and techniques for image pre-processing and augmentation;
4. Target condition: definition/classification and severity of glaucoma;
5. Reference standard: the clinical guidelines or diagnostic criteria upon which the primary study established the reference standard (e.g. numbers and qualification of human graders), diagnostic criteria/guidance/threshold values used for defining the presence of the target condition, if relevant;
6. Flow and timing: timing of taking images that were used as input for the reference standard(s) and for the index test(s); time intervals between taking images for those used as index tests, if more than one test is evaluated;
7. Comparative: threshold values or classification systems used by reference standards across included studies;
8. Others: source(s) of funding and conflicts of interest.

If performance results of more than one algorithm are reported in the primary studies, we will choose one algorithm per study, based on the best or optimal algorithm claimed by the study investigators in the conclusion of the primary study, so that each study appears once in the meta-analysis. When there is ambiguity in the conclusion of the primary study, we will extract data from the algorithm of the highest overall accuracy, sensitivity, or specificity, in order of preference.

When test performance data based on internal testing (or validation) and external validation are available in the same study, we will extract data from both evaluations, but include each in a separate meta-analysis as a planned subgroup analysis (see [Investigations of heterogeneity](#)).

Two review authors will compare the data collected and resolve any discrepancies. We will contact the study investigators if we need clarification or additional information.

Assessment of methodological quality

Two review authors will independently evaluate the risk of bias and applicability for each index test (AI algorithm) included.

For single test accuracy studies, we will use the revised Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) tool, outlined in Chapter 8 of the *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy* [25]. Based on the signalling questions provided in the tool, we will evaluate each study for four domains (participant selection, index test, reference standard, the flow and timing of tests) from each index test, and rate it as having a low, high, or unclear risk of bias.

For comparative accuracy studies, we will use the QUADAS-C tool, which is an extension of QUADAS-2, to assess risk of bias in studies comparing the accuracy of two or more tests [26]. First, we will

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evaluate the same four domains from QUADAS-2 for each index test, and then answer additional signalling questions for the pairwise comparisons.

We added two signalling questions to the QUADAS-2 checklist on topics that are not covered by the standard QUADAS-2 tool:

1. Were the performance results of algorithms properly (e.g. technique for internal validation used, such as resampling including cross-validation) validated when internal validation was used?
2. Were datasets split at the participant (or eye) level?

For both QUADAS-2 and QUADAS-C, we developed guidance, specific for this review, on how to answer each signalling question, and how to use the answers to reach an overall risk of bias judgement, and applicability for each domain (see [Supplementary material 2](#)).

Statistical analysis and data synthesis

We will carry out statistical analysis and data synthesis, outlined in Chapters 9 and 10 of the *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy* [27, 28].

We will extract or construct contingency (2 x 2) tables to show cross-classification of the index test compared with the standard reference test. The columns of these tables will reveal the true status (diseased versus not diseased) of the condition under investigation, while the rows show the index test results. From these tables, we will extract and calculate the sensitivities and specificities between the index test and the reference standard, with a 95% confidence interval.

We will use Review Manager (RevMan) to generate coupled forest plots [29], and receiver operating characteristic plots to visualise these calculations. We plan to fit hierarchical random-effects models to account for sensitivity, specificity, and the relationship between them; random sampling variability within studies; and heterogeneity in test accuracy between studies, including cases when there are too few studies or sparse data in the analysis [30]. The bivariate model estimates a summary point when studies apply the same threshold for test positivity, whereas the hierarchical summary receiver operating characteristic (HSROC) model estimates a summary receiver operating characteristic (SROC) curve when studies apply different thresholds for test positivity. Some studies may report sensitivity and specificity at more than one threshold, in which case, more complex analyses, using multiple thresholds per study, may provide more accurate estimates of SROC curves. We will choose the most suitable model, based on whether there is consensus amongst the included studies regarding the appropriate value for the positivity threshold. By consensus, we mean that the majority of the studies align at or around the same threshold value for positivity (i.e. testing positive for the target condition). Different AI algorithms might scale their probability outputs differently, so the same threshold value might not directly translate between models. Therefore, we will ensure that the studies agree on a standardised threshold value, or appropriately adjust for differences in scaling amongst the algorithms, when these threshold values for positivity are available.

To compare the accuracy of two or more tests, we will conduct direct (within-study) comparisons if a sufficient number of

comparative studies are available. Otherwise, we will undertake indirect comparisons, using data from all studies that evaluated the relevant tests.

We will investigate whether the observed test accuracy varies according to the clinical setting in which the test is used (namely, the general population in the community versus those referred to secondary care). If there is a sufficiently large number of studies for these two settings, we will report results for these two pathways in separate reviews.

If there are very few primary studies included in the review, then we will present a narrative summary to report the ranges of the reported sensitivity, specificity, and characteristics of the studies included, as outlined in Chapter 11 of the *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy* [31].

Investigations of heterogeneity

We will investigate heterogeneity, where possible, by adding covariates to the model, using the sources below, following the guidance in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy* [28]. We will perform these subgroup analyses when there are four or more studies in at least one subgroup, such that quantification of between-study heterogeneity is possible.

We will consider the following potential sources of heterogeneity in diagnostic performance in subgroup analysis.

1. Clinical setting in which the test is used (the general population in the community versus people referred to secondary care)
2. Study design (studies enrolling consecutive participants in the same setting versus multicentre registries and public databases)
3. Characteristics of the population (age according to quartiles, sex, symptomatic versus asymptomatic), when sufficient data are available
4. Prevalence of glaucoma in the training and test sets (< 5% versus ≥ 5%), since sensitivity and specificity may depend on disease prevalence [17]
5. Severity of glaucoma (including suspected glaucoma)
6. Reference standard (e.g. diagnosis based on functional deterioration versus structural changes versus both, numbers or speciality of human graders)
7. Core AI method used: neural networks (e.g. recurrent neural network (RNN), convolutional neural network (CNN), generative adversarial network (GAN), or others) versus other machine learning methods (e.g. random forest, or support vector machine)[32]
 - a. branched subgroup analysis by neural network structures, when feasible
8. Size of the dataset from which performance data were collected (< 1000 versus ≥ 1000 total unique participants)
9. Modalities of input data for the AI algorithms (imaging data, visual fields, clinical parameters, or any combination), particularly, as their accessibility and affordability may vary across different settings
10. Model development stage (internal validation versus external validation)

Sensitivity analyses

We will conduct a sensitivity analysis to determine how excluding index tests deemed at high overall risk of bias will affect the results of the primary analysis.

Assessment of reporting bias

For studies that have a study protocol, we will assess reporting biases by comparing prespecified outcomes (test performance metrics) described in the protocol with reported outcomes in the publication. For studies that do not have a study protocol, we will compare the methods with the results section.

Assessing the certainty of the evidence and summary of findings table

We will prepare a summary of findings table to present key study information, including the number of included studies, number of participants (eyes or images), the accuracy of the index test (sensitivity and specificity), the statistical uncertainty of the test accuracy measures reported, and the certainty of the evidence [33]. Two review authors will evaluate the certainty of the evidence according to the GRADE approach, by considering the five GRADE domains [34, 35].

1. Risk of bias
2. Indirectness (e.g. when there are important differences in the index tests studied versus those intended at the community or speciality clinic)[34]
3. Inconsistency
4. Imprecision
5. Publication bias

With the published guidance for diagnostic reviews [34, 35], we will clearly specify the rules for evaluating the certainty of evidence and pilot them on a subset of included studies, to ensure inter-rater reliability, before deploying them across all studies. We will categorise the certainty of a body of evidence as 'high', 'moderate', 'low', or 'very low'. We will resolve any discrepancies by discussion, or in consultation with a third review author. We will report the reasons for all decisions to downgrade the certainty of the evidence from high.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD016114](https://doi.org/10.1002/14651858.CD016114).

Supplementary material 1 Search strategies

Supplementary material 2 QUADAS-2/QUADAS-C Tool

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article:

- Sign-off Editor; clinical (final editorial decision): Tianjing Li, University of Colorado Anschutz Medical Campus, Denver, Colorado;
- Sign-off Editor; DTA (final editorial decision): Mariska M. G. Leeftang, Amsterdam UMC, University of Amsterdam;
- Sign-off Editor; DTA Trainee Editor (final editorial decision): Mohammad Yaghoobi, McMaster University and McMaster University Health Sciences Centre;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Anupa Shah, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Victoria Pennick, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Rafael Correia Barão, MD, FEBO, ULS Santa Maria, Lisbon, Portugal (clinical/content review); Dr. Denny Mathew John, Department of Community Medicine, Azeezia Institute of Medical Sciences and Research Kollam, Kerala, India (consumer review); Jérémie F Cohen, Université Paris Cité, Paris, France (methods review); Sarah Berhane, University of Birmingham (statistical review); Jo Platt, Central Editorial Information Specialist (search review).

Contributions of authors

Conceiving the protocol: MM, AA, SL, GV, KH

Designing the protocol: all

Writing the protocol: KV, RG, AS, SMN, SL, KH

Critical review of the protocol: all

Declarations of interest

KV: none

RG: none

AS: none

MM: received payment from SIFI (ophthalmic company) to attend meetings

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AA: none. AA is a Cochrane Editor, but was not involved in the editorial process for this protocol.

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GV: none. GV is a Cochrane Editor, but was not involved in the editorial process for this protocol.

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Registration and protocol

Cochrane approved the proposal for this review in January 2024.

Data, code and other materials

Data sharing is not applicable to this article as it is a protocol, so no datasets were generated or analysed.

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